# Agenda

1. Announcements
    - 12.1 and 12.2 HW due on Wednesday
    - Discussion Quiz 10 on Thursday over 12.1 and 12.2
    - Regrade Requests for Midterm 2 due Wednesday
    - Please fill out course evaluations
2. Finish 12.3
3. Start 12.4

**Activity:** Find the angle between the vectors

1.) $\langle 3,-1\rangle$ and $\langle -1,4\rangle$ 2.) $\langle -1,3,2\rangle$ and $\langle 4,2,-1\rangle$

① let $\langle 3,-1\rangle$ be $\vec{v}$ and $\vec{w}$ be $\langle -1,3,2\rangle$

$$\theta = \arccos\left(\frac{\vec{v}\cdot\vec{w}}{|\vec{v}||\vec{w}|}\right) = \arccos\left(\frac{-7}{\sqrt{3^2+(-1)^2}\sqrt{(-1)^2+4^2}}\right) = \arccos\left(\frac{-7}{\sqrt{10}\sqrt{16}}\right)$$

②

$$\theta = \arccos\left(\frac{-4+6+-2}{\sqrt{(-1)^2(3)^2(2)^2}\sqrt{(4)^2(2)^2(-1)^2}}\right) = \arccos(0) = \frac{\pi}{2}$$

perpendicular

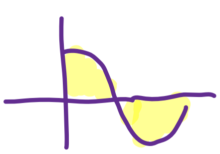
**Definition:** Two vectors $\vec{v}$ and $\vec{w}$ are orthogonal when the angle between them is $\pi/2$ $(90°)$

**Theorem:** Two vectors $\vec{v}$ + $\vec{w}$ are orthogonal if and only if $\vec{v}\cdot\vec{w} = 0$.

$$\vec{v}\cdot\vec{w} = |\vec{v}||\vec{w}|\cos\theta$$

**Note:**

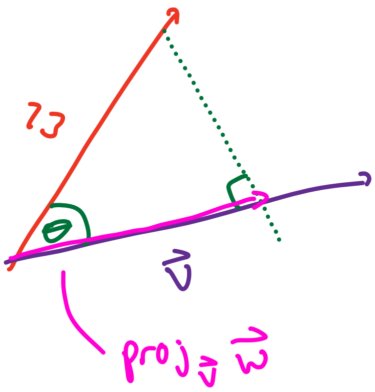
- If $\vec{v}\cdot\vec{w}$ then $\vec{v}$ and $\vec{w}$ are perpendicular/orthogonal with $\theta = \frac{\pi}{2}$



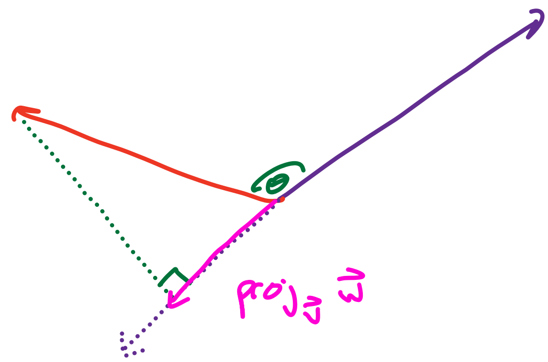
- If $\vec{v}\cdot\vec{w} > 0$, then $\theta$ is acute $(0 \le \theta < \frac{\pi}{2})$
- If $\vec{v}\cdot\vec{w} < 0$, then $\theta$ is obtuse $(0 < \theta \le \frac{\pi}{2})$
- If $\theta = 0$ or $\theta = \pi$, then $\vec{v}$ and $\vec{w}$ are parallel and $\vec{v} = k\vec{w}$

## Projections

acute:




obtuse angles of $\theta$:




"The projection of $\vec{w}$ on to $\vec{v}$."

$$\text{Proj}_{\vec{v}}\vec{w} = \underbrace{\left(\frac{\vec{v}\cdot\vec{w}}{|v|^2}\right)}_{\text{Scalar}}\vec{v} = \frac{|\vec{v}||\vec{w}|\cos\theta}{|\vec{v}|}\cdot\frac{\vec{v}}{|\vec{v}|}$$

$$= (|\vec{w}|\cos\theta)\underbrace{\left(\frac{1}{|\vec{v}|}\vec{v}\right)}_{\text{unit vector in same direction as } \vec{v}}$$

## Definition:

Let $\vec{v} \neq \vec{0}$ and $\vec{w}$ be vectors. Then the projection of $\vec{w}$ onto $\vec{v}$ is defined as

$$\text{Proj}_{\vec{v}}\vec{w} = \frac{\vec{v}\cdot\vec{w}}{|\vec{v}|^2}\vec{v} = \frac{(\vec{v}\cdot\vec{w})}{|\vec{v}|}\frac{1}{|\vec{v}|}\vec{v} = (\text{comp}_{\vec{v}}\vec{w})\left(\frac{1}{\vec{v}}\vec{v}\right)$$

unit vector

The scalar projection of $\vec{w}$ onto $\vec{v}$ is $\text{comp}_{\vec{v}}\vec{w} = \dfrac{\vec{v}\cdot\vec{w}}{|\vec{v}|}$

**Activity:** Let $\vec{w} = \langle -1, 2, 3\rangle$ and $\vec{v} = \langle 0, -1, 3\rangle$. Find $\text{proj}_{\vec{v}}\vec{w}$.

$$|\vec{v}|^2 = 0^2 + (-1)^2 + (3)^2 = 10 \qquad \vec{v}\cdot\vec{w} = (0)(-1) + (-1)(2) + (3)(3) = 7$$

$$\text{Proj}_{\vec{v}}\vec{w} = \frac{7}{10}\langle 0, -1, 3\rangle = \langle 0, \frac{7}{10}, \quad \rangle$$

# The Cross Product

Let $\vec{v} = \langle x_1, y_1, z_1 \rangle$ and $\vec{w} = \langle x_2, y_2, z_2 \rangle$ be vectors in $\mathbb{R}^3$. The cross product is:

way 1

$$\vec{v} \times \vec{w} = \langle y_1 z_2 - y_2 z_1 ,\ x_2 z_1 - x_1 z_2 ,\ x_1 y_2 - x_2 y_1 \rangle$$

way 2

determinate

$$\vec{v} \times \vec{w} = \begin{array}{c} \begin{array}{ccc} + & - & + \end{array} \\ \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} \end{array} \begin{array}{l} \leftarrow \text{standard basis vectors} \\ \leftarrow \vec{v} \\ \leftarrow \vec{w} \end{array}$$

$$= \begin{vmatrix} y_1 & z_1 \\ y_2 & z_2 \end{vmatrix} \vec{i} - \begin{vmatrix} x_1 & z_1 \\ x_2 & z_2 \end{vmatrix} \vec{j} + \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} \vec{k}$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$= (y_1 z_2 - y_2 z_1) i - (x_1 z_2 - x_2 z_1)\vec{j} + (x_1 y_2 - x_2 y_1)\vec{k}$$

**Example:** Let $\vec{v} = \langle 2, 1, -1 \rangle$ and $\vec{w} = \langle -3, 4, 1 \rangle$. Find $\vec{v} \times \vec{w}$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -1 \\ -3 & 4 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 4 & 1 \end{vmatrix} i - \begin{vmatrix} 2 & -1 \\ -3 & 1 \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & 1 \\ -3 & 4 \end{vmatrix} \vec{k}$$

$$= (1-(-4))\vec{i} - (2-3)\vec{j} + (8-(-3))$$

$$= 5i - \vec{j} + 11\vec{k} = \langle 5, 1, 11 \rangle$$

Activity: Let $\vec{v} = \langle 2, 1, -1 \rangle$ and $\vec{w} = \langle -3, 4, 1 \rangle$. Find $\vec{w} \times \vec{v}$

## Properties of the Cross Product

Theorem

Theorem:

**Properties** Let $\vec{v}$, $\vec{u}$, and $\vec{w}$ be vectors in $\mathbb{R}^3$.

Let $c$ be a constant.

1.) 2.)

3.) 4.)

5.) 6.)

7.)

**Theorem:** Two vectors $\vec{v}$ and $\vec{w}$ are parallel if and only if $\vec{v} \times \vec{w} = \vec{0}$.